

Extending Imperfect Retraining Analysis to Three-Component MLS (Continue)

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I. INTRODUCTION

In the previous meeting, two areas were highlighted for further investigation before proceeding with the three-component MLS extension. The first concerned *policy complexity*: the base paper [1] analyzes only two retraining policies—progressive and conservative—but does not examine how many possible policies exist for the two-component system, nor where these two policies sit within that larger space. Understanding this combinatorial structure is a necessary foundation before extending the analysis to three components, where the policy space grows by orders of magnitude.

The second area concerned *evaluation metrics*: specifically, how to measure performance degradation in the planning module (C3) of the autonomous driving pipeline. Unlike perception and prediction, where ground-truth comparisons are straightforward, planning raises fundamental difficulties—there is no unique correct plan, degradation can be silent in simple scenarios, and poor system output may reflect upstream errors rather than planner failure. Without clear metrics, the binary state signals required by the SMP model cannot be defined.

This report presents the research conducted on both topics. Section II formalizes the policy complexity of the two-component MLS, enumerating the full decision space and identifying unexplored policy families. Section III proposes a three-tier evaluation framework for the planning module that resolves the metric definition challenge and maps directly to the SMP state representation.

II. POLICY COMPLEXITY ANALYSIS FOR TWO-COMPONENT MLS

The base paper [1] proposes two retraining policies—progressive and conservative—but does not ask a prior question: *how many retraining policies are possible, and where do these two sit within that space?* This section answers that question step by step.

A. System States

The two-component MLS has an upstream model u and a downstream model d . Each model either meets its performance threshold (1) or does not (0), producing four possible system states:

Because the system output is determined by the downstream model d , the system is *available* whenever d meets its threshold—that is, in states $(1, 1)$ and $(0, 1)$ —and *unavailable* otherwise.

TABLE I
SYSTEM STATES AND THEIR INTERPRETATIONS

State	System available?	Interpretation
$(1, 1)$	Yes	Both models meet threshold
$(0, 1)$	Yes	u degraded, but d still meets threshold
$(1, 0)$	No	d fails; system output is unavailable
$(0, 0)$	No	Both models fail

B. Retraining Actions

At any state, the system operator chooses one of the following actions:

TABLE II
AVAILABLE RETRAINING ACTIONS

Action	Meaning
$\emptyset$	Do nothing (wait)
$\{u\}$	Retrain only the upstream model
$\{d\}$	Retrain only the downstream model
$\{u, d\}$	Retrain both models

Formally, let $\mathcal{C} = \{u, d\}$ be the component set. The action set is the power set $\mathcal{A} = \mathcal{P}(\mathcal{C}) = \{\emptyset, \{u\}, \{d\}, \{u, d\}\}$, giving $|\mathcal{A}| = 4$ options at each state.

C. What Is a Retraining Policy?

A *deterministic retraining policy* is a rule that assigns exactly one action to each state. Formally, it is a function:

$$\pi : \mathcal{S} \rightarrow \mathcal{A}, \quad (1)$$

where $\mathcal{S} = \{(1, 1), (0, 1), (1, 0), (0, 0)\}$. A single policy can be written as a four-entry table. For example:

If state is...	then do...
$(1, 1)$	$\emptyset$
$(0, 1)$	$\{u\}$
$(1, 0)$	$\{d\}$
$(0, 0)$	$\{u, d\}$

Since there are 4 states and 4 choices per state, the total number of distinct deterministic policies is:

$$|\Pi| = |\mathcal{A}|^{|\mathcal{S}|} = 4^4 = 256. \quad (2)$$

D. Pruning Unreasonable Policies

Not all 256 policies are operationally sensible. Two domain-informed constraints reduce the count.

Constraint 1. At state $(1, 1)$, both models work. There is no reason to retrain, so the only sensible action is “do nothing”:

$$\pi(1, 1) = \emptyset. \quad (\text{reduces choices from 4 to 1}) \quad (3)$$

Constraint 2. At state $(0, 0)$, the system is down and both models have failed. Doing nothing serves no purpose, so at least one component must be retrained:

$$\pi(0, 0) \neq \emptyset. \quad (\text{reduces choices from 4 to 3}) \quad (4)$$

At the two mixed states, $(0, 1)$ and $(1, 0)$, every action—including “do nothing”—can be justified by a valid operational argument. All 4 options remain.

The number of *reasonable* policies is therefore:

$$|\Pi_{\text{constrained}}| = \underbrace{1}_{(1,1)} \times \underbrace{4}_{(0,1)} \times \underbrace{4}_{(1,0)} \times \underbrace{3}_{(0,0)} = 48. \quad (5)$$

E. Where the Base Paper’s Two Policies Sit

The progressive policy π_P and conservative policy π_C are two specific entries in this 48-policy space:

TABLE III
SIDE-BY-SIDE COMPARISON OF THE TWO BASE PAPER POLICIES

State	Progressive π_P	Conservative π_C
$(1, 1)$	$\emptyset$	$\emptyset$
$(0, 1)$	$\{u\}$	$\emptyset$
$(1, 0)$	$\{d\}$	$\{d\}$
$(0, 0)$	$\{u, d\}$	$\{u, d\}$

The two policies are identical at three of the four states. They differ at exactly one state— $(0, 1)$ —where progressive retrains the upstream model proactively while conservative waits. The base paper’s entire analysis therefore rests on a **single binary decision** within a space of 48 reasonable policies.

F. Unexplored Policy Families

The remaining 46 policies include several structurally distinct families that the base paper does not consider. We highlight four, each embodying a different retraining philosophy.

1) *Downstream-First Policy* (π_{DF}): At state $(0, 0)$, retrain only d instead of both components:

$$\pi_{DF} = (\emptyset, \{u\}, \{d\}, \{d\}). \quad (6)$$

If d recovers, the system reaches $(0, 1)$ without ever risking Type 2 entangled enhancement from concurrent upstream retraining. The upstream model is addressed only after d is restored. The trade-off is a longer expected recovery time from state $(0, 0)$.

2) *Joint Retraining Policy* (π_J): Retrain both components whenever anything is degraded:

$$\pi_J = (\emptyset, \{u, d\}, \{u, d\}, \{u, d\}). \quad (7)$$

This is the most aggressive approach: it maximizes recovery rate by always retraining everything. The cost is doubled computational overhead and potential entanglement at states where only one component has degraded.

3) *Upstream-Avoidant Policy* (π_{UA}): Never retrain u in isolation:

$$\pi_{UA} = (\emptyset, \emptyset, \{d\}, \{d\}). \quad (8)$$

Entangled enhancement arises specifically when u is retrained and the resulting representation shift degrades d . By never retraining u alone, this policy eliminates both Type 1 and Type 2 entangled enhancement by construction. The cost is that u may remain degraded indefinitely, eventually harming availability when dataset shift worsens.

4) *Threshold-Conditional Policy* (π_{TC}): Introduce a tunable parameter $\gamma \in [0, 1]$ that governs the decision at state $(0, 1)$:

$$\pi_{TC}(0, 1) = \begin{cases} \{u\}, & \text{if } c_1 \geq \gamma, \\ \emptyset, & \text{otherwise,} \end{cases} \quad (9)$$

where c_1 is the coverage factor (probability of successful upstream recovery without triggering entangled enhancement). Setting $\gamma = 0$ recovers the progressive policy; $\gamma = 1$ recovers the conservative policy. This family reveals that the base paper’s two policies are endpoints of a continuous spectrum, not two fundamentally different strategies.

G. Summary and Implications

Table IV collects all named policies.

TABLE IV
POLICY TAXONOMY: ACTIONS AT THE TWO CONTESTED STATES

Policy	$\pi(0, 1)$	$\pi(0, 0)$	Key property
Progressive π_P	$\{u\}$	$\{u, d\}$	Proactive upstream fix
Conservative π_C	$\emptyset$	$\{u, d\}$	Reactive only
Downstream-first π_{DF}	$\{u\}$	$\{d\}$	Avoids c_2 risk
Joint π_J	$\{u, d\}$	$\{u, d\}$	Maximum aggression
Upstream-avoidant π_{UA}	$\emptyset$	$\{d\}$	Eliminates entanglement

The base paper’s availability-continuity trade-off is derived from comparing 2 policies in a space of at least 48. The trade-off frontier it identifies may therefore not be the true Pareto frontier: there may exist policies that improve one metric without sacrificing the other.

This observation becomes critical for three-component systems. With $N = 3$ components, the number of states grows to $2^3 = 8$, the number of actions to $2^3 = 8$, and the unconstrained policy count to $8^8 \approx 1.68 \times 10^7$. Exhaustive enumeration is infeasible at that scale, making the structural approach developed here—grouping policies into families defined by qualitative properties—a necessary foundation for the three-component extension.

TABLE V
POLICY COMPLEXITY SCALING WITH COMPONENT COUNT

Components N	States $ \mathcal{S} $	Actions $ \mathcal{A} $	Policies $ \Pi $
2	4	4	256
3	8	8	$\approx 1.68 \times 10^7$
4	16	16	$\approx 1.84 \times 10^{19}$

III. EVALUATION METRICS FOR THREE-COMPONENT MLS

A prerequisite for the SMP model is a binary signal for each component: working (1) or degraded (0). In the base paper’s two-component system, the metrics were straightforward—disparity MAE for the upstream model and AP_R40 for the downstream model [?]. In our three-component autonomous driving pipeline, defining this signal for the planning module (C3) is substantially harder. This section explains why, and proposes a three-tier evaluation framework that resolves the difficulty.

A. Why Perception and Prediction Are Easy to Evaluate

C1 (Perception) detects objects from sensor data. The nuScenes dataset provides ground-truth bounding boxes for every annotated frame [2]. Evaluation reduces to a direct comparison: did C1 find the objects that actually exist? The standard metric is **mAP** (mean Average Precision). If mAP falls below a threshold τ_1 , C1 is degraded.

C2 (Prediction) forecasts the future trajectories of detected agents. nuScenes provides ground-truth future positions at 2-second and 6-second horizons. Evaluation again reduces to comparison: did C2 predict where the agent actually went? The standard metrics are **minADE** (minimum Average Displacement Error) and **minFDE** (minimum Final Displacement Error), both measured in meters. If minADE exceeds a threshold τ_2 , C2 is degraded.

In both cases, a single well-established metric, compared against annotated ground truth, produces a clear degradation signal.

B. Why Planning Is Fundamentally Harder to Evaluate

C3 (Planning) generates the ego-vehicle’s driving trajectory. Three properties make its evaluation qualitatively different from C1 and C2.

1) *No unique ground truth*: For perception, either there is a car at position X or there is not. For planning, multiple trajectories may be equally valid. The human driver’s recorded path is one acceptable plan, but not the only one. A planner that takes a slightly different but equally safe route is not degraded—yet any displacement-based metric would penalize it.

2) *Plans can succeed locally but fail globally*: Suppose C1 fails to detect a pedestrian. C2 therefore does not predict the pedestrian’s movement. C3 produces a trajectory that drives through the pedestrian’s location. From C3’s perspective the plan is optimal—there is nothing in the way. The system-level

output is catastrophic, but *C3 itself has not degraded*. The fault lies upstream. A naïve evaluation of C3’s output cannot distinguish this case from genuine C3 degradation.

3) *Degradation can be silent*: On a straight, empty highway, even a severely degraded planner will output “go straight,” which happens to be correct. The degradation only becomes visible when C3 faces a complex scenario. A single-metric threshold may fail to detect degradation in benign conditions.

C. Proposed Three-Tier Evaluation Framework

To address these challenges, we propose evaluating C3 at three tiers: plan quality, component isolation, and inter-component coupling. Together, these tiers provide the signals required by the SMP model.

1) *Tier 1: Plan Quality Metrics*: These measure the output of C3 directly, regardless of input source.

TABLE VI
PLAN QUALITY METRICS FOR C3

Metric	What it measures
L2 displacement error	Distance between planned trajectory and human driver’s trajectory at fixed horizons (1 s, 2 s, 3 s)
Collision rate	Whether the planned path collides with any agent (binary per scenario)
Drivable area compliance	Whether the planned path stays within drivable road surface
Comfort (jerk, lateral acceleration)	Whether the trajectory is physically smooth or erratic

L2 displacement error is the primary metric, consistent with the nuScenes planning benchmark. Collision rate is the primary safety metric. Drivable area compliance and comfort metrics serve as secondary indicators that can detect abnormal planner behavior even when displacement error appears acceptable.

2) *Tier 2: Component-Isolated Evaluation*: Plan quality metrics alone cannot distinguish between “C3 is degraded” and “C3 is fine but receiving bad inputs from C1 or C2.” To resolve this, we evaluate C3 under two conditions:

nosep

- **Isolated evaluation**: Feed C3 ground-truth detections and ground-truth predictions from nuScenes annotations, bypassing C1 and C2 entirely. Measure plan quality.
- **Pipeline evaluation**: Feed C3 the actual outputs of C1 and C2 in the live pipeline. Measure plan quality.

Comparing the two reveals the source of any degradation:

This isolated evaluation is what determines the binary state s_3 in the SMP independently of upstream components. The same principle applies to C2: feeding C2 ground-truth detections (bypassing C1) isolates C2’s own performance and determines s_2 .

TABLE VII
DIAGNOSING DEGRADATION SOURCE VIA ISOLATED VS. PIPELINE
EVALUATION

C3 isolated	C3 pipeline	Diagnosis
Good	Good	System is working
Good	Bad	C3 is fine; problem is upstream (C1 or C2)
Bad	Bad	C3 itself is degraded; retrain C3

3) *Tier 3: Inter-Component Coupling Metrics*: The third tier quantifies how strongly upstream errors amplify through the pipeline. We define the **error propagation ratio** as:

$$\rho_{i \rightarrow j} = \frac{\Delta M_j / M_j}{\Delta M_i / M_i}, \quad (10)$$

where M_i is the performance metric of component C_i , and ΔM_i is the change in that metric after a retraining event or dataset shift. For example, if C1’s mAP drops by 5% and C3’s L2 error increases by 20%, then $\rho_{1 \rightarrow 3} = 4$, indicating strong coupling.

This ratio serves two purposes in the SMP framework: noseop

- It characterizes how likely entangled enhancement is: high ρ means small upstream changes cause large downstream effects, making entanglement more probable.
- It informs the coverage factors $c_1, c_2, \dots$ in the SMP transition model, which can be calibrated from experimentally measured propagation ratios.

D. Mapping Metrics to SMP States

The three-tier framework produces the signals needed for the 8-state SMP. Table VIII summarizes how each component state and the system availability are determined.

TABLE VIII
METRIC-TO-STATE MAPPING FOR THE THREE-COMPONENT SMP

Signal	What is evaluated	Metric	Condition
s_1	C1 vs. ground-truth annotations	mAP	$\text{mAP} \geq \tau_1$
s_2	C2 with ground-truth detections as input	minADE	$\text{minADE} \leq \tau_2$
s_3	C3 with ground-truth predictions as input	L2 error	$\text{L2} \leq \tau_3$
System	Full pipeline C1→C2→C3 end-to-end	L2 error	$\text{L2} \leq \tau_{\text{sys}}$

The distinction between component state and system availability is critical. A component can pass its own threshold ($s_3 = 1$) while the system is unavailable because upstream components are degraded. Conversely, all components can individually meet their thresholds while the system still fails due to coupling effects. This gap between component-level and system-level performance is precisely where entangled enhancement occurs, and it is this gap that the SMP model is designed to capture.

IV. CONCLUSION

Extending the imperfect retraining analysis to a three-component MLS requires systematically addressing both policy complexity and component evaluation metrics. This report

formalizes the policy space, demonstrating that previously analyzed strategies are merely endpoints within a broader space of 48 viable policies—a combinatorial reality that is critical to understand before scaling. Additionally, we proposed a robust three-tier evaluation framework specifically tailored for the autonomous driving planning module. By isolating individual component performance and quantifying inter-component coupling, this framework effectively yields the binary degradation signals necessary to construct an accurate SMP state representation.

REFERENCES

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